WEB TECHNOLOGY LAB MANUAL

HTML, CSS3 & JavaScript Practical Exercises (1 to 5)

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Experiment 1: Familiarization of HTML and CSS Basics

Aim

To study the basic structure of an HTML document and understand fundamental CSS styling properties such as colors, fonts, borders, and the box model.

Theory / Description

HTML (HyperText Markup Language) is the standard markup language used to create the structure and content of web pages. It consists of elements represented by tags such as <h1>, <p>, <ul>, <table>, and <a>.

CSS (Cascading Style Sheets) is used to control the presentation of HTML elements, including colors, fonts, spacing, borders, and layout. CSS can be applied inline, internally within a <style> tag, or externally via a linked .css file.

This lab covers text formatting tags, ordered and unordered lists, tables, hyperlinks, and the CSS box model (margin, border, padding, content).

Procedure (Step-by-Step)

1. Create a new file named index.html.
2. Add the basic HTML5 boilerplate: <!DOCTYPE html>, <html>, <head>, and <body> tags.
3. Inside <head>, add a <style> block to define CSS rules for headings, paragraphs, tables, and links.
4. In the <body>, add headings, formatted text (bold, italic, underline), ordered/unordered lists, a table, a hyperlink, and a bordered <div>.
5. Save the file and open it in a web browser to view the rendered output.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Lab 1 - HTML & CSS Basics</title>
<style>
  body {
    font-family: Arial, Helvetica, sans-serif;
    background-color: #f4f6f8;
    margin: 0;
    padding: 20px;
    color: #222;
  }
  h1 {
    color: #ffffff;
    background-color: #2c3e50;
    padding: 12px 20px;
    border-radius: 6px;
  }
  h2 {
    color: #2c3e50;
    border-bottom: 2px solid #2c3e50;
    padding-bottom: 4px;
  }
  p {
    line-height: 1.6;
  }
  .highlight {
    background-color: #fff3cd;
    padding: 2px 6px;
    border-radius: 3px;
  }
  ul li {
    margin-bottom: 6px;
  }
  table {
    border-collapse: collapse;
    width: 60%;
```

```
    margin-top: 10px;
  }
  table, th, td {
    border: 1px solid #999;
  }
  th, td {
    padding: 8px 12px;
    text-align: left;
  }
  th {
    background-color: #2c3e50;
    color: white;
  }
  a {
    color: #2980b9;
    text-decoration: none;
    font-weight: bold;
  }
  .box {
    border: 2px dashed #2980b9;
    padding: 15px;
    width: 300px;
    margin-top: 15px;
  }
</style>
</head>
<body>

  <h1>HTML &amp; CSS Basics Demo</h1>

  <h2>1. Text Formatting</h2>
  <p>This paragraph demonstrates <b>bold</b>, <i>italic</i>, <u>underline</u> and a
  <span class="highlight">highlighted</span> word using inline CSS classes.</p>

  <h2>2. Lists</h2>
  <ul>
    <li>Unordered list item one</li>
    <li>Unordered list item two</li>
    <li>Unordered list item three</li>
  </ul>
  <ol>
    <li>Ordered step one</li>
    <li>Ordered step two</li>
  </ol>

  <h2>3. Table</h2>
  <table>
    <tr><th>Tag</th><th>Purpose</th></tr>
    <tr><td>&lt;h1&gt;-&lt;h6&gt;</td><td>Headings</td></tr>
    <tr><td>&lt;p&gt;</td><td>Paragraph</td></tr>
    <tr><td>&lt;table&gt;</td><td>Tabular data</td></tr>
  </table>

  <h2>4. Link and Box (CSS Box Model)</h2>
  <p><a href="#">This is a styled hyperlink</a></p>
  <div class="box">This div has a border, padding, and margin applied through CSS &mdash; illustrating the
CSS box model.</div>

</body>
</html>
```

Sample Output

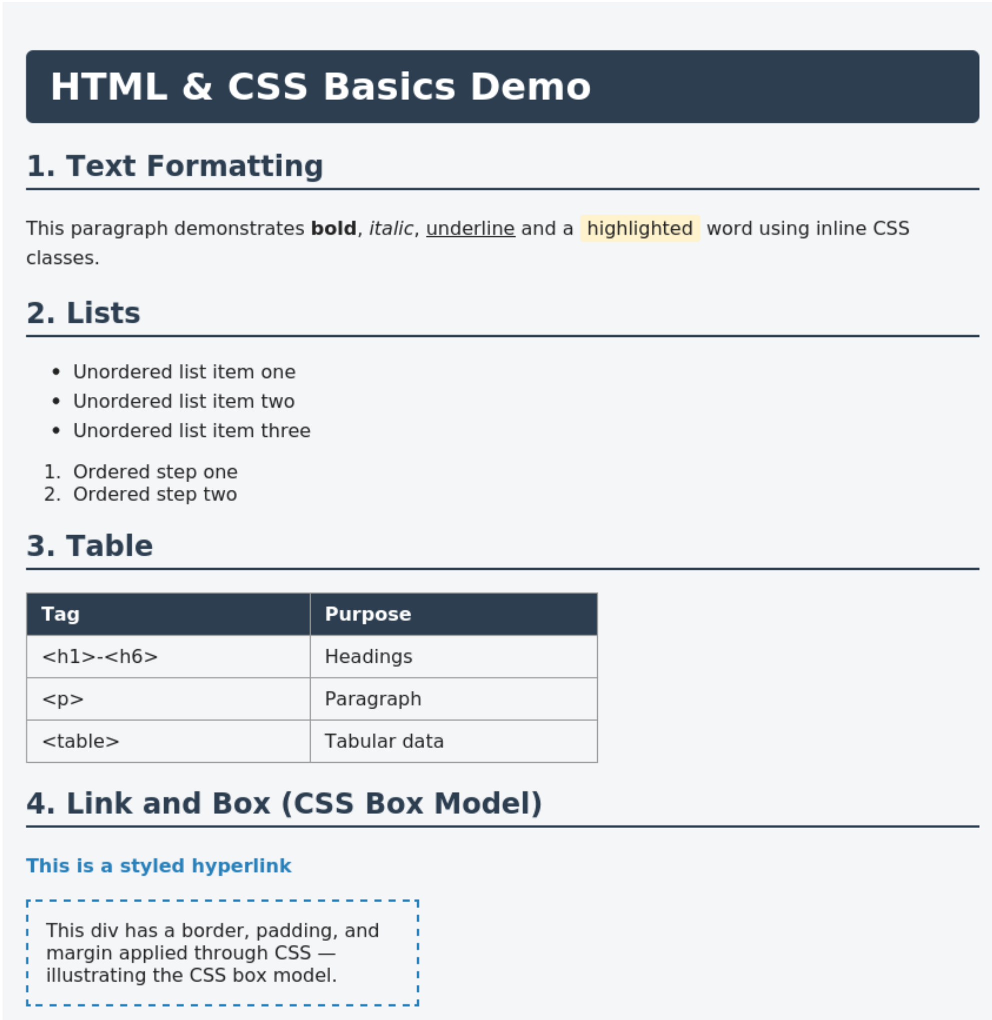

Fig. 1: Browser output of Experiment 1

Experiment 2: HTML Webpage Showcasing Biodata with CSS Styling

Aim

To design a biodata (personal information) webpage using HTML for structure and CSS for styling, including layout, colors, and a card-based design.

Theory / Description

A biodata page typically presents personal, educational, and other details in an organised, readable format. HTML tables or CSS Flexbox/Grid can be used to align label-value pairs neatly.

This exercise uses CSS properties such as linear-gradient() backgrounds, box-shadow, border-radius, and Flexbox (display: flex) to create a modern 'card' style layout, along with tables for structured data display.

Procedure (Step-by-Step)

6. Create index.html and define the page structure with a container <div class="card"> holding a header and content section.
7. In the header, add a circular photo placeholder and the person's name and title, styled using border-radius: 50% and Flexbox alignment.
8. In the content area, use HTML tables to display Personal Details, Education, and Hobbies sections.
9. Apply CSS for background gradient, box-shadow on the card, and consistent spacing/typography.
10. Save and open the file in a browser to verify the biodata layout renders correctly.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Lab 2 - Biodata</title>
<style>
  body {
    font-family: 'Segoe UI', Arial, sans-serif;
    background: linear-gradient(135deg, #e0eafc, #cfdef3);
    display: flex;
    justify-content: center;
    padding: 30px;
    margin: 0;
  }
  .card {
    background: #ffffff;
    width: 650px;
    border-radius: 10px;
    box-shadow: 0 6px 18px rgba(0,0,0,0.15);
    overflow: hidden;
  }
  .header {
    background-color: #34495e;
    color: white;
    padding: 20px 25px;
    display: flex;
    align-items: center;
    gap: 20px;
  }
  .photo {
    width: 90px;
    height: 90px;
    border-radius: 50%;
    background: #7f8c8d;
    display: flex;
    align-items: center;
    justify-content: center;
    font-size: 32px;
    border: 3px solid white;
  }
```

```
    .header h1 {
      margin: 0;
      font-size: 24px;
    }
    .header p {
      margin: 4px 0 0;
      color: #dcdde1;
    }
    .content {
      padding: 20px 25px;
    }
    table {
      width: 100%;
      border-collapse: collapse;
    }
    table td {
      padding: 8px 6px;
      border-bottom: 1px solid #ecf0f1;
    }
    table td.label {
      font-weight: bold;
      color: #34495e;
      width: 35%;
    }
    .section-title {
      margin-top: 18px;
      color: #34495e;
      border-left: 4px solid #2980b9;
      padding-left: 8px;
    }
  </style>
  </head>
  <body>
    <div class="card">
      <div class="header">
        <div class="photo">RS</div>
        <div>
          <h1>Rohan Sharma</h1>
          <p>Aspiring Software Engineer</p>
        </div>
      </div>
      <div class="content">
        <h2 class="section-title">Personal Details</h2>
        <table>
          <tr><td class="label">Date of Birth</td><td>12th March 2004</td></tr>
          <tr><td class="label">Gender</td><td>Male</td></tr>
          <tr><td class="label">Nationality</td><td>Indian</td></tr>
          <tr><td class="label">Address</td><td>Bengaluru, Karnataka, India</td></tr>
          <tr><td class="label">Email</td><td>rohan.sharma@example.com</td></tr>
          <tr><td class="label">Phone</td><td>+91-98765-43210</td></tr>
        </table>

        <h2 class="section-title">Education</h2>
        <table>
          <tr><td class="label">Degree</td><td>B.E. Computer Science (Pursuing)</td></tr>
          <tr><td class="label">College</td><td>ABC Institute of Technology</td></tr>
          <tr><td class="label">CGPA</td><td>8.7 / 10</td></tr>
        </table>

        <h2 class="section-title">Hobbies</h2>
        <table>
          <tr><td>Reading, Coding, Cricket, Photography</td></tr>
        </table>
      </div>
    </div>
  </body>
  </html>
```

Sample Output

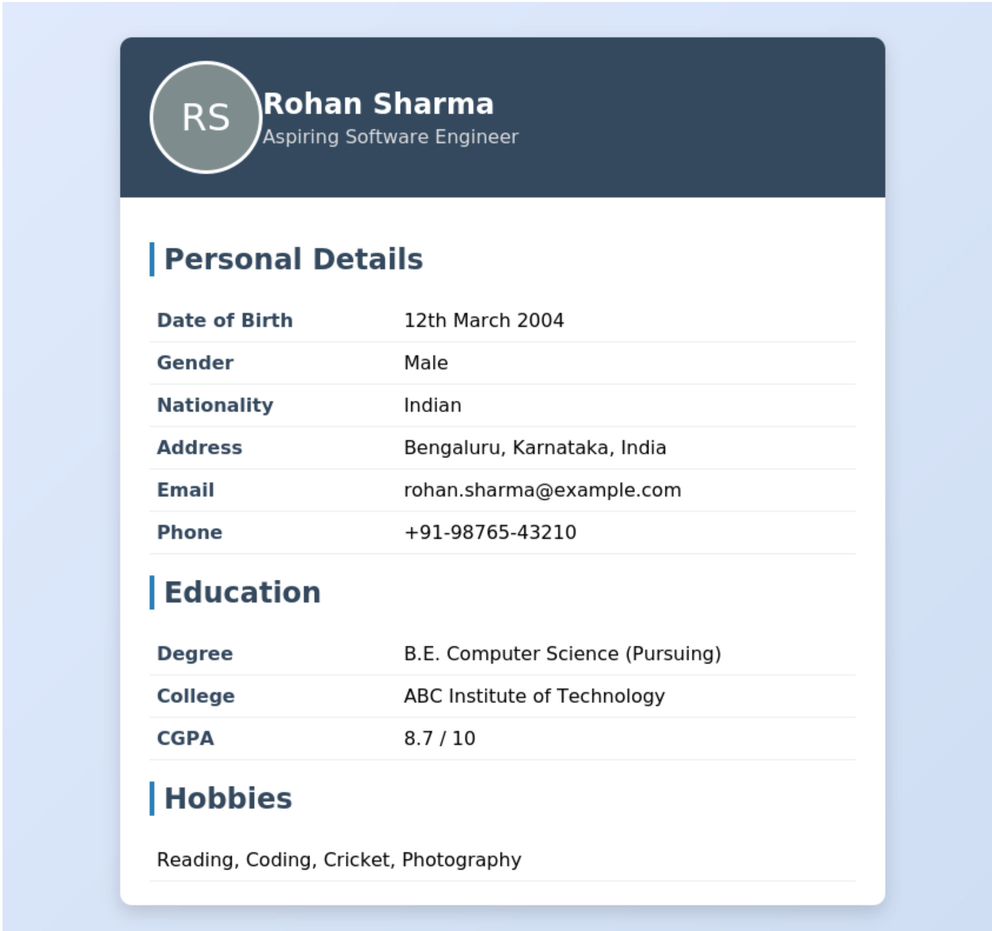

Fig. 2: Browser output of Experiment 2

Experiment 3: Interactive Webpage for a New Restaurant Using CSS3 Features

Aim

To design an interactive restaurant webpage using CSS3 features such as gradients, Flexbox/Grid layout, transitions, hover effects, and keyframe animations.

Theory / Description

CSS3 introduced advanced styling capabilities beyond basic CSS, including linear-gradient() and radial-gradient() for backgrounds, CSS Grid (display: grid) for two-dimensional layouts, transition for smooth property changes, and @keyframes for animations.

Interactivity in this lab is achieved purely with CSS (no JavaScript) using the :hover pseudo-class to create hover-lift effects on menu cards, and an infinite pulse animation on the call-to-action button using @keyframes.

Procedure (Step-by-Step)

11. Create index.html with a <header> containing the restaurant name and tagline, styled with a gradient background.
12. Add a <nav> bar with navigation links; style links with a color transition on hover.
13. Create a menu section using CSS Grid (grid-template-columns: repeat(3, 1fr)) to display dish cards in three columns.
14. For each dish card, apply a hover effect using transform: translateY() and scale() combined with a transition for smoothness.
15. Add a 'Book a Table' button with a pulsing box-shadow animation defined using @keyframes.
16. Save and preview the page in a browser; hover over the menu cards to observe the interactive lift effect.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Lab 3 - Spice Garden Restaurant</title>
<style>
  * { box-sizing: border-box; }
  body {
    font-family: 'Georgia', serif;
    margin: 0;
    background: #fdfaf5;
  }
  header {
    background: linear-gradient(135deg, #8e2de2, #c0392b);
    color: white;
    text-align: center;
    padding: 50px 20px;
    text-shadow: 1px 1px 3px rgba(0,0,0,0.4);
  }
  header h1 {
    margin: 0;
    font-size: 42px;
    letter-spacing: 2px;
  }
  header p {
    font-style: italic;
    margin-top: 8px;
  }
  nav {
    background: #2c3e50;
    display: flex;
    justify-content: center;
    gap: 30px;
    padding: 12px;
  }
  nav a {
```

```
      color: white;
      text-decoration: none;
      font-weight: bold;
      transition: color 0.3s ease;
    }
    nav a:hover {
      color: #f1c40f;
    }
    .menu {
      display: grid;
      grid-template-columns: repeat(3, 1fr);
      gap: 20px;
      padding: 40px;
    }
    .dish {
      background: white;
      border-radius: 12px;
      box-shadow: 0 4px 10px rgba(0,0,0,0.12);
      padding: 20px;
      text-align: center;
      transition: transform 0.3s ease, box-shadow 0.3s ease;
    }
    .dish:hover {
      transform: translateY(-8px) scale(1.03);
      box-shadow: 0 10px 20px rgba(0,0,0,0.2);
    }
    .dish .icon {
      font-size: 48px;
    }
    .dish h3 {
      color: #c0392b;
      margin: 10px 0 5px;
    }
    .price {
      font-weight: bold;
      color: #27ae60;
    }
    .cta {
      text-align: center;
      padding: 30px;
    }
    .btn {
      background: #c0392b;
      color: white;
      padding: 14px 34px;
      border: none;
      border-radius: 30px;
      font-size: 16px;
      cursor: pointer;
      box-shadow: 0 4px 12px rgba(192,57,43,0.4);
      animation: pulse 2s infinite;
    }
    @keyframes pulse {
      0% { box-shadow: 0 0 0 0 rgba(192,57,43,0.5); }
      70% { box-shadow: 0 0 0 15px rgba(192,57,43,0); }
      100% { box-shadow: 0 0 0 0 rgba(192,57,43,0); }
    }
    footer {
      background: #2c3e50;
      color: #bdc3c7;
      text-align: center;
      padding: 15px;
      font-size: 14px;
    }
  </style>
  </head>
  <body>
    <header>
      <h1>Spice Garden</h1>
      <p>Authentic Flavours, Modern Ambience</p>
    </header>
    <nav>
```

```
      <a href="#">Home</a>
      <a href="#">Menu</a>
      <a href="#">About</a>
      <a href="#">Reservations</a>
      <a href="#">Contact</a>
    </nav>

    <section class="menu">
      <div class="dish">
        <div class="icon">🍛</div>
        <h3>Paneer Butter Masala</h3>
        <p>Rich and creamy North Indian curry.</p>
        <p class="price">₹220</p>
      </div>
      <div class="dish">
        <div class="icon">🍜</div>
        <h3>Hakka Noodles</h3>
        <p>Wok-tossed noodles with veggies.</p>
        <p class="price">₹180</p>
      </div>
      <div class="dish">
        <div class="icon">🍕</div>
        <h3>Farmhouse Pizza</h3>
        <p>Loaded with fresh garden vegetables.</p>
        <p class="price">₹250</p>
      </div>
    </section>

    <div class="cta">
      <button class="btn">Book a Table</button>
    </div>

    <footer>&copy; 2026 Spice Garden Restaurant. All rights reserved.</footer>
  </body>
  </html>
```

Sample Output

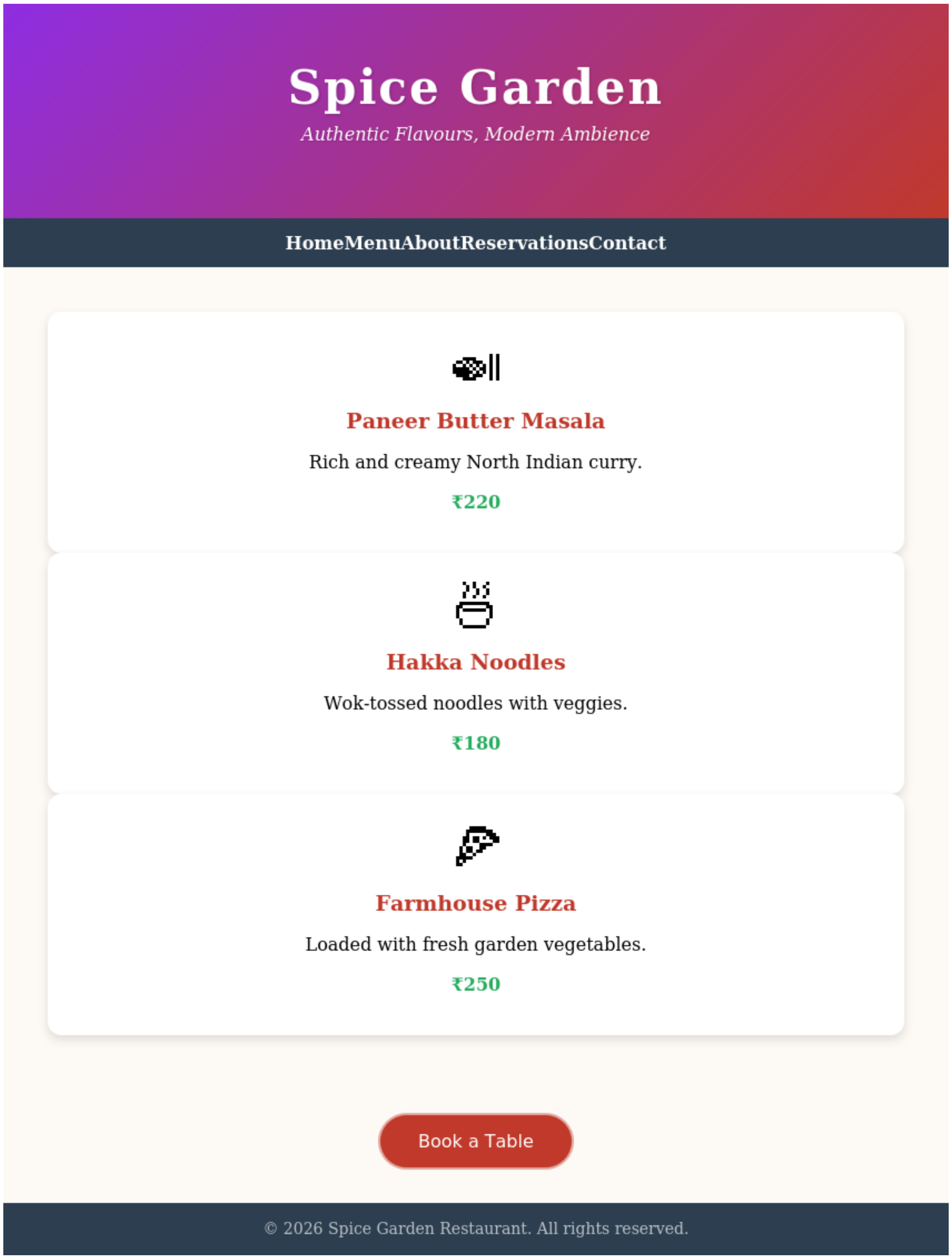

Fig. 3: Browser output of Experiment 3

Experiment 4: Simple Web Form to Gather User Information

Aim

To create an HTML form using various input types to collect user information, styled using CSS for a clean and usable layout.

Theory / Description

HTML forms use the <form> element along with input controls such as <input type="text">, type="email", type="tel", type="date", <select>, <textarea>, radio buttons, and checkboxes to collect different kinds of user data.

The <label> element is associated with an input using the for attribute (matching the input's id) to improve accessibility and usability.

CSS is used here to style form width, spacing, borders, and the submit button, giving the form a clean card-like appearance.

Procedure (Step-by-Step)

17. Create index.html and add a <form> element containing a heading.
18. Add labelled text, email, tel, and date input fields for Name, Email, Phone, and Date of Birth respectively.
19. Add a set of radio buttons for Gender and a <select> dropdown for City.
20. Add checkboxes for Interests and a <textarea> for additional comments.
21. Add a Submit button and style all elements using CSS for consistent spacing and appearance.
22. Save and open the file in a browser; fill in sample values to test the form fields.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Lab 4 - User Information Form</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background: #eef2f5;
    display: flex;
    justify-content: center;
    padding: 30px;
    margin: 0;
  }
  form {
    background: white;
    width: 450px;
    padding: 25px 30px;
    border-radius: 8px;
    box-shadow: 0 4px 12px rgba(0,0,0,0.1);
  }
  h2 {
    text-align: center;
    color: #2c3e50;
    margin-top: 0;
  }
  label {
    display: block;
    margin-top: 14px;
    margin-bottom: 6px;
    font-weight: bold;
    color: #34495e;
  }
  input[type="text"],
  input[type="email"],
  input[type="tel"],
  input[type="date"],
  select,
  textarea {
```

```
      width: 100%;
      padding: 8px 10px;
      border: 1px solid #ccc;
      border-radius: 5px;
      font-size: 14px;
    }
    .radio-group, .checkbox-group {
      display: flex;
      gap: 15px;
      margin-top: 4px;
    }
    button {
      margin-top: 20px;
      width: 100%;
      background: #2980b9;
      color: white;
      padding: 10px;
      border: none;
      border-radius: 5px;
      font-size: 16px;
      cursor: pointer;
    }
    button:hover {
      background: #216390;
    }
  </style>
  </head>
  <body>
    <form>
      <h2>User Registration Form</h2>

      <label for="name">Full Name</label>
      <input type="text" id="name" placeholder="Enter your full name" required>

      <label for="email">Email Address</label>
      <input type="email" id="email" placeholder="you@example.com" required>

      <label for="phone">Phone Number</label>
      <input type="tel" id="phone" placeholder="+91-XXXXXXXXXX">

      <label for="dob">Date of Birth</label>
      <input type="date" id="dob">

      <label>Gender</label>
      <div class="radio-group">
        <label><input type="radio" name="gender"> Male</label>
        <label><input type="radio" name="gender"> Female</label>
        <label><input type="radio" name="gender"> Other</label>
      </div>

      <label for="city">City</label>
      <select id="city">
        <option>Bengaluru</option>
        <option>Mumbai</option>
        <option>Delhi</option>
        <option>Chennai</option>
      </select>

      <label>Interests</label>
      <div class="checkbox-group">
        <label><input type="checkbox"> Sports</label>
        <label><input type="checkbox"> Music</label>
        <label><input type="checkbox"> Reading</label>
      </div>

      <label for="msg">Message</label>
      <textarea id="msg" rows="3" placeholder="Any additional comments"></textarea>

      <button type="submit">Submit</button>
    </form>
```

```
</body>
</html>
```

Sample Output

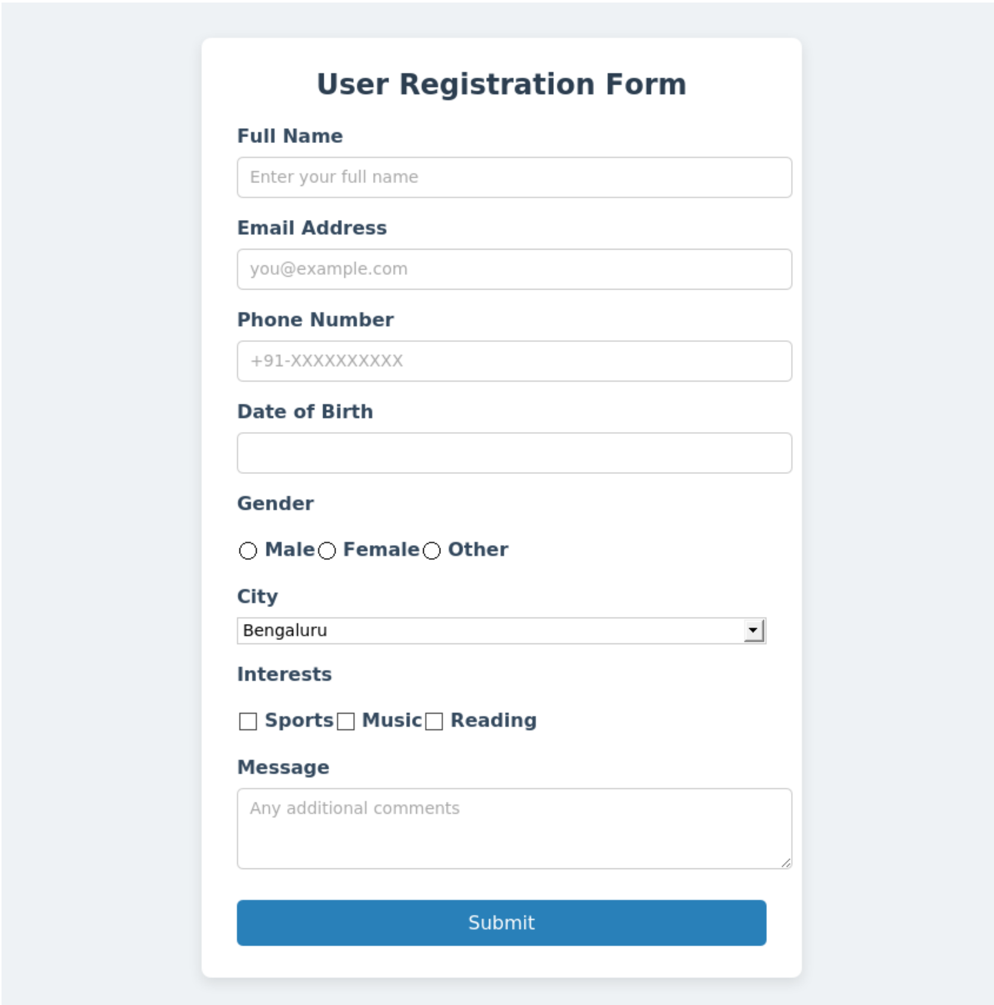

Fig. 4: Browser output of Experiment 4

Experiment 5: Basic JavaScript Exercises Including a Canvas Drawing Application

Aim

To practice fundamental JavaScript concepts such as variables, functions, and DOM manipulation, and to build an interactive canvas-based drawing application using the HTML5 Canvas API.

Theory / Description

JavaScript is a client-side scripting language used to add interactivity to web pages. Functions defined in JavaScript can be triggered by HTML events such as onclick.

DOM (Document Object Model) manipulation allows JavaScript to read and modify the content and style of HTML elements at runtime using methods like document.getElementById().

The HTML5 <canvas> element provides a drawable region on a web page. Using its 2D rendering context (canvas.getContext('2d')), JavaScript can draw lines, shapes, and text, and respond to mouse events (mousedown, mousemove, mouseup) to build a freehand drawing application.

Procedure (Step-by-Step)

23. Create index.html with sections for: (a) sum of two numbers, (b) even/odd checker, (c) DOM text/style manipulation, and (d) a canvas drawing app.
24. Write JavaScript functions addNumbers() and checkEvenOdd() that read input values using document.getElementById().value and display results in a <div>.
25. Write a changeText() function that modifies the innerText and CSS style of a paragraph when a button is clicked.
26. Add a <canvas> element and obtain its 2D context using getContext('2d').
27. Attach mousedown, mousemove, and mouseup event listeners to the canvas to implement freehand drawing, using ctx.beginPath(), ctx.lineTo(), and ctx.stroke().
28. Add a color picker and range slider to control stroke colour and brush size, and a Clear button using ctx.clearRect().
29. Save the file and test all four exercises in a browser, drawing freehand on the canvas with the mouse.

Source Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Lab 5 - JavaScript Exercises & Canvas Drawing App</title>
<style>
  body {
    font-family: Arial, sans-serif;
    background: #f4f6f8;
    padding: 25px;
    margin: 0;
  }
  h1 { color: #2c3e50; }
  section {
    background: white;
    padding: 18px 22px;
    border-radius: 8px;
    margin-bottom: 20px;
    box-shadow: 0 2px 6px rgba(0,0,0,0.08);
  }
  h2 { color: #2980b9; margin-top: 0; }
  button {
    background: #2980b9;
    color: white;
    border: none;
    padding: 8px 16px;
    border-radius: 5px;
    cursor: pointer;
    margin-right: 8px;
```

```
    }
    #result, #evenOddResult {
      font-weight: bold;
      margin-top: 8px;
      color: #27ae60;
    }
    canvas {
      border: 2px solid #34495e;
      background: white;
      cursor: crosshair;
      display: block;
      margin-top: 10px;
    }
    .toolbar {
      margin-bottom: 8px;
    }
    .toolbar input[type="color"] {
      width: 40px;
      height: 30px;
      vertical-align: middle;
    }
    .toolbar input[type="range"] {
      vertical-align: middle;
    }
  </style>
  </head>
  <body>
    <h1>JavaScript Practical Exercises</h1>

    <section>
      <h2>Exercise 1: Sum of Two Numbers</h2>
      <input type="number" id="num1" value="12" style="width:70px">
      +
      <input type="number" id="num2" value="8" style="width:70px">
      <button onclick="addNumbers()">Calculate</button>
      <div id="result"></div>
    </section>

    <section>
      <h2>Exercise 2: Check Even or Odd</h2>
      <input type="number" id="numCheck" value="7" style="width:70px">
      <button onclick="checkEvenOdd()">Check</button>
      <div id="evenOddResult"></div>
    </section>

    <section>
      <h2>Exercise 3: DOM Manipulation - Change Text &amp; Color</h2>
      <p id="domText">This text will change on button click.</p>
      <button onclick="changeText()">Click Me</button>
    </section>

    <section>
      <h2>Exercise 4: Canvas Drawing Application</h2>
      <div class="toolbar">
        Color: <input type="color" id="colorPicker" value="#e74c3c">
        Brush Size: <input type="range" id="brushSize" min="1" max="20" value="4">
        <button onclick="clearCanvas()">Clear</button>
      </div>
      <canvas id="drawCanvas" width="500" height="300"></canvas>
    </section>

  <script>
    // Exercise 1
    function addNumbers() {
      const a = parseFloat(document.getElementById('num1').value);
      const b = parseFloat(document.getElementById('num2').value);
      document.getElementById('result').innerText = "Sum = " + (a + b);
    }

    // Exercise 2
    function checkEvenOdd() {
```

```
      const n = parseInt(document.getElementById('numCheck').value);
      const msg = (n % 2 === 0) ? n + " is Even" : n + " is Odd";
      document.getElementById('evenOddResult').innerText = msg;
    }

    // Exercise 3
    function changeText() {
      const el = document.getElementById('domText');
      el.innerText = "Text changed dynamically using JavaScript DOM manipulation!";
      el.style.color = "#c0392b";
      el.style.fontWeight = "bold";
    }

    // Exercise 4 - Canvas Drawing App
    const canvas = document.getElementById('drawCanvas');
    const ctx = canvas.getContext('2d');
    let drawing = false;

    canvas.addEventListener('mousedown', (e) => {
      drawing = true;
      ctx.beginPath();
      ctx.moveTo(e.offsetX, e.offsetY);
    });
    canvas.addEventListener('mousemove', (e) => {
      if (!drawing) return;
      ctx.lineWidth = document.getElementById('brushSize').value;
      ctx.lineCap = 'round';
      ctx.strokeStyle = document.getElementById('colorPicker').value;
      ctx.lineTo(e.offsetX, e.offsetY);
      ctx.stroke();
    });
    window.addEventListener('mouseup', () => drawing = false);

    function clearCanvas() {
      ctx.clearRect(0, 0, canvas.width, canvas.height);
    }

    // Demo: auto-draw a sample sketch on load to illustrate the app
    window.onload = function() {
      addNumbers();
      checkEvenOdd();
      ctx.lineWidth = 4;
      ctx.lineCap = 'round';
      ctx.strokeStyle = '#e74c3c';
      ctx.beginPath();
      ctx.moveTo(50, 150);
      ctx.bezierCurveTo(150, 30, 250, 270, 350, 150);
      ctx.stroke();

      ctx.strokeStyle = '#2980b9';
      ctx.beginPath();
      ctx.arc(420, 80, 35, 0, Math.PI * 2);
      ctx.stroke();

      ctx.fillStyle = '#27ae60';
      ctx.font = '20px Arial';
      ctx.fillText('Sample Sketch', 170, 260);
    };
  </script>
  </body>
  </html>
```

Sample Output

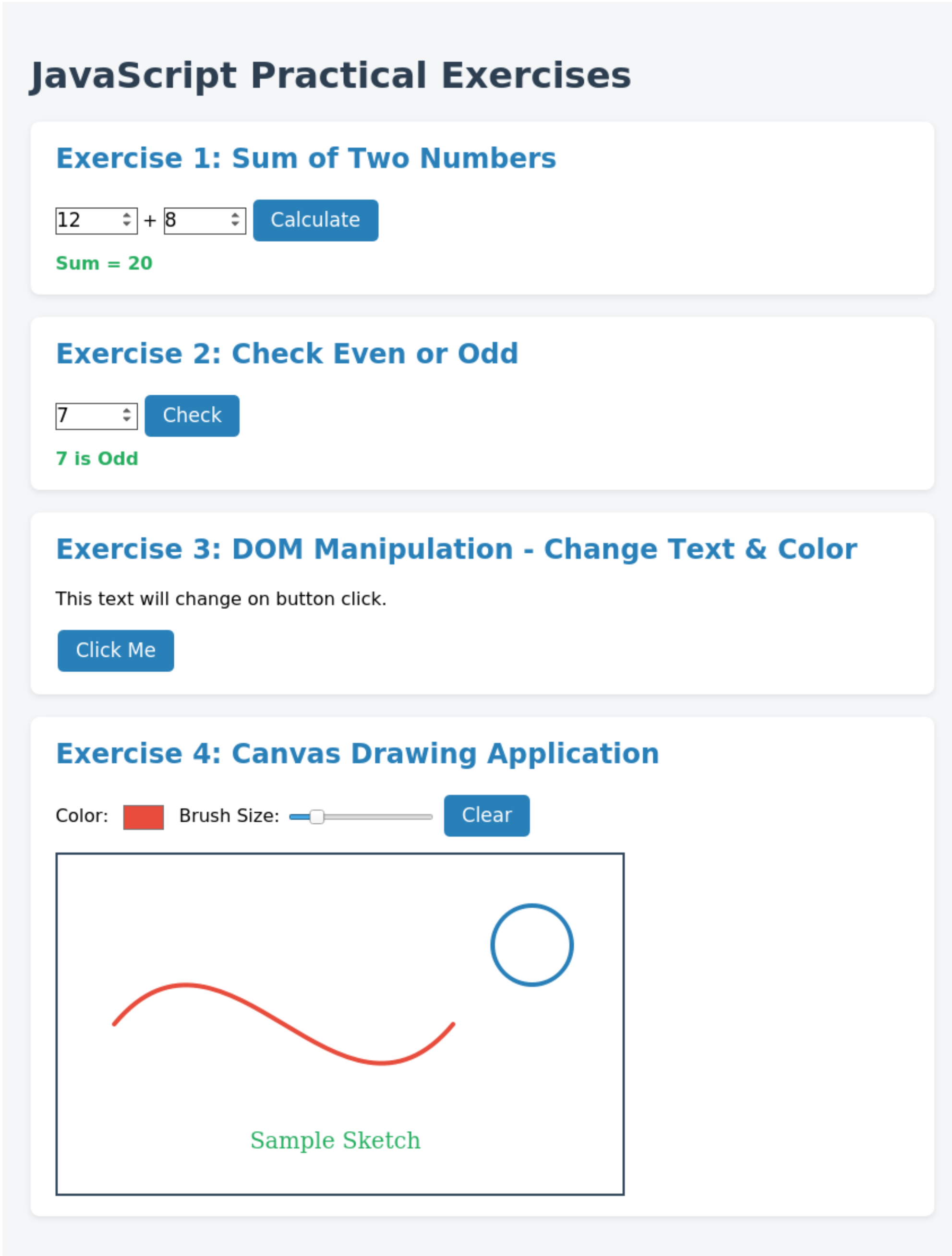

Fig. 5: Browser output of Experiment 5